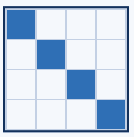
MATRIX · MULTIPLICATION

Four ways to read the same product

Every row computes AB and every row gives the same answer. What differs is the unit being assembled — a scalar, a column, a row, or a whole rank-one matrix — and each unit makes a different fact obvious.

4
READINGS

ONE NUMBER AT A TIME — 1

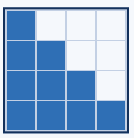


Entry view | the definition

ASSEMBLED FROM	FORMULA	NATURAL WHERE
a single scalar $(AB)_{ij}$	$(AB)_{ij} = \sum_k a_{ik}b_{kj}$	hand computation, proofs

Row i of A dotted with column j of B. Correct and unilluminating — it says how to get a number out but not what the product **is**, which is why the three readings below exist.

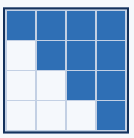
ONE VECTOR AT A TIME — 2



Column view | read B column by column

ASSEMBLED FROM	FORMULA	NATURAL WHERE
a column of AB	column j of $AB = A\mathbf{b}_j$	column space, transformations

Each column of the product is A applied to the corresponding column of B — so every column of AB is a linear combination of the columns of A. This is why $\text{Col}(AB) \subseteq \text{Col}(A)$, and hence why the rank of a product is bounded.

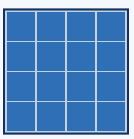


Row view | read A row by row

ASSEMBLED FROM	FORMULA	NATURAL WHERE
a row of AB	row i of $AB = \mathbf{a}_i B$	row space, elimination

The mirror statement: every row of AB is a combination of the rows of B. This is the reading that explains why row operations — which are left multiplications — preserve the row space.

ONE MATRIX AT A TIME — 1



Outer product sum | pair column k of A with row k of B

ASSEMBLED FROM	FORMULA	NATURAL WHERE
a rank-one matrix	$AB = \sum_k \mathbf{a}_k \mathbf{b}_k^T$	SVD, low-rank approximation

The product as a sum of rank-one pieces. Truncating that sum is exactly what low-rank approximation does, which makes this the reading that matters most in practice and the one taught last.

The definition is the entry view, but it is the one that explains least. Each of the other three is the same sum regrouped, and each is the natural reading somewhere else in the subject — the column view for linear combinations, the outer product for low-rank approximation.